Object Oriented Programming (CSEG1044)

Unit 04 – Lecture 02: Generic Methods + Bounds + Wildcards

April 1, 2026

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Quick Links

Overview Key Topics Demo Summary

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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Lecture Snapshot

- **Unit focus:** Generics, Lambdas, Swing, and JDBC

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Lecture Snapshot

- **Unit focus:** Generics, Lambdas, Swing, and JDBC
- **Course thread:** Use Generic Methods + Bounds + Wildcards to move Campus Resource Manager toward a real desktop application with UI, callbacks, and persistence.

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Recall Before You Start

- **Generic classes**: This lecture extends the first generics lecture from type declarations to more flexible operations. Main reference: `Unit 04 / Lecture 01 slides/notes`.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Recall Before You Start

- **Generic classes**: This lecture extends the first generics lecture from type declarations to more flexible operations. Main reference: `Unit 04 / Lecture 01 slides/notes`.
- **Inheritance**: Bounds and wildcards only make sense if subtype relationships are already familiar. Main reference: `Unit 02 / Lecture 04 slides/notes`.

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Learning Outcomes

- Write a generic method and explain why it reduces duplication.

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Learning Outcomes

- Write a generic method and explain why it reduces duplication.
- Use bounds (`extends`) to restrict allowed types.

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Learning Outcomes

- Write a generic method and explain why it reduces duplication.
- Use bounds (`extends`) to restrict allowed types.
- Use wildcards (`? extends`, `? super`) with collections.

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Key Terms to Know

- **Generic method**: method with its own type parameter, e.g., `static <T> T first(List<T> list)`.
- **Bound**: restriction on type parameter, e.g., `<T extends Number>`.
- **Wildcard**: unknown type parameter, e.g., `List<?>`.
- **Upper-bounded wildcard**: `? extends Type` (read/produce).
- **Lower-bounded wildcard**: `? super Type` (write/consume).
- **Invariance**: `List<Integer>` is not a subtype of `List<Number>`.

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Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.

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Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.
2. Study the main build in this order: *Generic methods* → *Bounds (restrict allowed types)* → *Wildcards: making collection parameters flexible*.

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3. Trace the worked example and connect each line back to one concept frame.

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Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.
2. Study the main build in this order: *Generic methods* → *Bounds (restrict allowed types)* → *Wildcards: making collection parameters flexible*.
3. Trace the worked example and connect each line back to one concept frame.
4. Attempt the mini exercises before moving to the recap and exit check.

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Unit 04: Generics, Lambdas, GUI Swing & Database Connectivity

- Lecture focus: Generic Methods + Bounds + Wildcards

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Today: Roadmap

- Generic methods

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Today: Roadmap

- Generic methods
- Bounded type parameters and wildcards (intro level)

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Today: Roadmap

- Generic methods
- Bounded type parameters and wildcards (intro level)
- Using generics with collections (patterns that show up in real code)

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Generic method (idea)

- Make only the method generic (not necessarily the class).

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Generic method (idea)

- Make only the method generic (not necessarily the class).
- Syntax: `public static <T> void printArray(T[] a)`

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Bounds (restrict types)

- `<T extends Number>` limits `T` to numeric types.

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Bounds (restrict types)

- `<T extends Number>` limits `T` to numeric types.
- Common comparison pattern: `<T extends Comparable<T>»`

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Why wildcards? (invariance)

- `List<Integer>` is **not** a `List<Number>`.

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Why wildcards? (invariance)

- `List<Integer>` is **not** a `List<Number>`.
- We need flexible method parameters for collections.

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Wildcards: `extends` vs `super`

- `? extends T`: use when list produces/read values of `T`.

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Wildcards: `extends` vs `super`

- `? extends T`: use when list produces/read values of `T`.
- `? super T`: use when list consumes/you add values of `T`.

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Wildcards: `extends` vs `super`

- `? extends T`: use when list produces/read values of `T`.
- `? super T`: use when list consumes/you add values of `T`.
- PECS: Producer Extends, Consumer Super.

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Checkpoint and Reflection

Reflection prompt

Where would *Generic Methods + Bounds + Wildcards* matter most in Campus Resource Manager, and which design choice would you justify using this lecture's ideas?

Quick self-check

Which part needs one more example before you move on: *Generic methods*, *Bounds (restrict allowed types)*, the worked example, or the recap?

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Example: generic method

```
public static <T> void printArray(T[] a) {
    for (T x : a) System.out.print(x + " ");
    System.out.println();
}
```

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Example: bounded max

```
public static <T extends Comparable<T>> T max(T a, T b) {
    return (a.compareTo(b) >= 0) ? a : b;
}
```

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Example: wildcard sum

```
import java.util.List;
public static double sum(List<? extends Number> list) {
    double s = 0;
    for (Number n : list) s += n.doubleValue();
    return s;
}
```

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Mini Exercises (2–3 mins)

1. Why is `List<Integer>` not a `List<Number>`?

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Mini Exercises (2–3 mins)

1. Why is `List<Integer>` not a `List<Number>`?
2. Which wildcard for reading: `extends` or `super`?

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Mini Exercises (2–3 mins)

1. Why is `List<Integer>` not a `List<Number>`?
2. Which wildcard for reading: `extends` or `super`?
3. State the PECS rule.

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Watch-outs

- Using a wildcard when a named type parameter would be clearer.

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Watch-outs

- Using a wildcard when a named type parameter would be clearer.
- Overengineering beginner APIs with too many bounds.

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Watch-outs

- Using a wildcard when a named type parameter would be clearer.
- Overengineering beginner APIs with too many bounds.
- Assuming List<Child> is automatically a List<Parent>.

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Key Takeaways

- Generic methods reduce duplication.

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Key Takeaways

- Generic methods reduce duplication.
- Bounds restrict allowed types (safer APIs).

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Key Takeaways

- Generic methods reduce duplication.
- Bounds restrict allowed types (safer APIs).
- Wildcards make collection parameters flexible (PECS).

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Checkpoint Questions

1. Write one example of a generic method signature.

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Checkpoint Questions

1. Write one example of a generic method signature.
2. What does `<T extends Number>` restrict?

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Checkpoint Questions

1. Write one example of a generic method signature.
2. What does `<T extends Number>` restrict?
3. When do you use `? super T`?

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Next Study Steps

- Revisit the recall references if any older idea still feels weak.

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- Rework the lecture example without looking at the notes first.

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- Rework the lecture example without looking at the notes first.
- Attempt the self-study practice and then answer the exit questions in full sentences.
- Apply the idea once inside Campus Resource Manager to make the concept stick.

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Next Lecture Preview

Next: Lambdas + Functional Interfaces

- We will write short callback functions and use them with collections and UI events.

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